Solution Requirements

SMARTBRIDGE - Solution Requirements

Date: 21/09/2026

Team ID: SWTID-2026-4223

Team Size: 4

Team Leader: Sweatha P

Team Members:Lakshmi Narayanan R, Rithika P, Sandhiya K

Project Name: AI FitTrack

Maximum Marks: 4 Marks

Step 1: Functional Requirements (FR)

1. Authentication — Users register using email and password. Passwords are encrypted using bcryptjs. Login issues a signed JWT token. Priority: High.

2. Authorization levels — Authenticated users can manage their personal workouts and invoke AI features. Middleware restricts unauthorized API calls. Priority: High.

3. External interfaces — Integrates with Google Gemini AI API using the @google/genai SDK for workout generation and analytics. Priority: High.

4. Transactions processing — Handles transactional CRUD requests for workout records (create, read, search, update, delete) returning JSON responses. Priority: High.

5. Reporting — Generates performance reports including total workouts, average duration, and total calories burned via AI insights. Priority: Medium.

6. Business rules, etc. — Validation rules: Workout duration must be > 0, calories burned ≥ 0, and emails must be unique. Priority: High.

7. Compliance to laws — Ensures user privacy by salting/hashing passwords and avoiding exposure of credentials in API payloads. Priority: High.

8. Other — Full-text query searching for workouts based on workout name, category, or workout date. Priority: Medium.

Step 2: Non-Functional Requirements (NFR)

1. Performance & Speed — Fast response times for CRUD APIs and Gemini AI request processing. Target: API responses < 200ms; AI generation < 3 seconds.

2. Scalability — Handles growing database records and concurrent user requests. Target: MongoDB Atlas cluster scales seamlessly up to 10,000 users.

3. Security & Data Privacy — Secure transmission, encrypted stored passwords, and protected API endpoints. Target: Passwords hashed with bcrypt (salt 10); JWT authentication required.

4. Reliability & Availability — High server uptime and resilient database connectivity. Target: 99.9% uptime per month with automatic MongoDB Atlas connection retry.

5. Usability & Accessibility — Standard RESTful API output structure and clear error responses. Target: 100% standardized JSON format with status codes (200, 201, 400, 500).

6. Maintainability — Clean Code separation following the MVC architecture. Target: Modular separation across Controllers, Services, Models, and Routes.